

Research paper

Exergetic, economic and exergy-based sustainability analysis of a power generation system with CO₂ capture and methanol production

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ABSTRACT

This study proposes a new biomass-based Combined Heat and Power (CHP)/renewable methanol production through captured carbon dioxide. A liquid organic hydrogen carrier (LOHC) stores and releases hydrogen. Literature review shows that LOHC has been integrated into such a system for the first time. The exergetic, exergoeconomic, and economic performance are researched. On average, results show that the exergetic efficiency of the system is 6.6 %. Depletion ratio (DR) is the depletion rate of the exergy source given to the system, which is desired to have low values. The sustainability index (SI) is the DR's reverse, representing the required exergy source per exergy destruction (exergy depletion). High values of this index indicate less exergy source consumed in the system. Exergetic ecological index (EECI) compares the useful products or depleted exergy per consumed exergy source destruction. The negative value of EECI means that the depletion of the exergy source is more significant than the product exergy and vice versa. The exergy-based sustainability indices, including DR, SI, and EECI, are found as 0.934, 1.071, and -0.868 , respectively. Exergoeconomic analysis reveals that the exergy destruction rate at the system's capital cost (R_{EX}) value is 0.087×10^2 . Heat Recovery Steam Generator (HRSG), gasifier, and Heat Exchanger 1 (HE1) are obtained as 5.153×10^{-2} (kW/€), 1.707×10^{-2} (kW/€), 3.094×10^{-2} (kW/€) respectively. The payback period and the R_{EX} are 14.56 years and 0.087×10^{-2} (kW/€), respectively; it seems long for an engineering application, but these technologies are relatively new, and their costs might decrease soon. Exergy destruction or lost power per capita, which means depleted investment, costs much less than 1. Although the thermodynamic values of the system seem low because irreversibilities occurred in chemical reactions, CO₂ saved and produced methanol and cannot be ignored in terms of its environmental and economic benefits, considering CO₂ taxes and the methanol economy.

1. Introduction

The environmental issues resulting from harmful emissions, especially CO₂, require urgent solutions. CO₂ is mostly released by the energy generation systems fueled by fossil fuels. That's why renewable energy applications have been focused for a few decades. Biomass is one of the most important renewable energy sources (Tursi and Olivito, 2021; Paraschiv and Paraschiv, 2023). Bioenergy can be used by direct combustion of the bioresources or converted into biogas or syngas via

fermentation and gasification (Wang et al., 2023). In the last decade, many scientists have claimed that biomass energy is not carbon-neutral as previously thought in terms of rapid combustion, as G. Ahamer summarized in (Ahamer, 2022) or explained by the IEA Bioenergy program in (IEA, 2024). It means that, in theory, CO₂ is theoretically absorbed by trees for a long time, and the combustion of biomass resources in a limited time could increase emissions just at the moment when the world needs to sharply reduce emissions and reach goals of becoming net-zero by 2050. For this reason, CO₂ capture from biomass has been researched (Zhang et al., 2022). Another problem with the CO₂

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Nomenclature		p	pressure
c	specific cost (€/unit) or heat capacity (J/K)	P	Product
C	Cost (€)	phys	physical
$\dot{E}x$	exergy rate (kW)	syg	syngas
h	specific enthalpy o (kJ/kg)	w	water
H	enthalpy (kJ)	<i>Greek symbols</i>	
I	interest rate (-)	η_{dehyd}	dehydrogenation efficiency
In	inflation rate (-)	η_C	isentropic efficiency of the compressor
k	isentropic coefficient	Δ	ratio hydrogen stored in the LOHC system
n	mole number (-)	ε	specific chemical exergy (kJ/kmol)
$\dot{m}$	flow rate (kg/s)	λ	exergetic coefficient for a substance
M	molecular weight (kg/kmol)	φ	exergy efficiency
q	reaction heat (K)	β	heat exchanger effectiveness
$\dot{Q}$	heat rate (kW)	<i>Abbreviations</i>	
r	compression ratio	AC	air compressor
R	universal gas constant (J/molK)	CC	combustion chamber
$R_{EX,k}$	exergy destruction at capital cost for component k (€/kW)	CCS	CO2 capture system
s	specific entropy (kJ/kgK)	CHP	combined heat and power
S	entropy (kJ/kgK)	DHYD	dehydrogenation reactor
T	temperature (K)	DR	depletion ratio
$\dot{W}$	power (kW)	EECI	exergetic ecological index
x	mole fraction (-)	EES	engineering equation solver software
Z_k	capital investment of the k component (€)	EXCEM	exergy-cost-energy-mass
<i>Subscripts</i>		FGC	feed gas compressor
B	biomass	GT	gas turbine
chem	chemical	HE	heat exchanger
Comp	compressor	HSRG	heat recovery steam generation
dehyd	dehydrogenation	HYD	hydrogenation reactor
E	electricity	LHV	lower heating value
F	fuel	LOHC	liquid organic hydrogen carrier
hyd	hydrogenation	MEA	monoethanolamine
i	i th species	MTH	methanization reactor
in	input	OMC	operation and maintenance cost
IM	income	PH	preheater
LOHC_	hydrogenated LOHC	PP	payback period
LOHC+	dehydrogenated LOHC	PSA	pressure swing adsorber
M	methanol	SI	sustainability index
o	environment	ST	steam turbine
out	output	Syg Comp	syngas compressor

is “what will we do with this CO₂?” and the cost of the system with the CO₂ Capture System (CCS). Generating electricity from biomass and selling it to the grid with CCS can provide extra income by avoiding CO₂. However, there is still a problem with using the captured CO₂ by CCS since storing the CO₂ generates an added cost (Middleton and Yaw, 2018; Mechleri et al., 2017). For all this, converting captured CO₂ into methanol using renewable hydrogen has been considered a promising new method (Sollai et al., 2023a; Zhang et al., 2024). Renewable hydrogen will be obtained from the syngas produced by biomass. This process can be achieved using liquid organic hydrogen carriers (LOHC). The advantage of the LOHC is that hydrogen can be absorbed from the syngas, as seen in refs (Jorschick et al., 2018; Açikkalp et al., 2023). and can be stored under environmental conditions. The disadvantage of this process is the reduction of the syngas’ calorific value due to hydrogen absorption; however, it can be compensated by increasing syngas production.

Since LOHCs are promising solutions for efficient and safe energy storage, many studies have recently been conducted on them. These include studies on developing potential and ideal LOHC systems with existing LOHC systems (Cho et al., 2021; Rao and Yoon, 2020; Sekine and Higo, 2021). Chemical and economic assessments of the LOHC

system were studied by Niermann et al (Niermann et al., 2019a). They concluded that LOHCs have a significant potential for energy transition. Niermann et al. made a techno-economic assessment for a defined process chain (Niermann et al., 2019b). They considered different LOHCs and evaluated their performance in terms of storage and transportation conditions. They stated that LOHCs have the potential for energy transport; however, some improvements in reactors, process heat integration, and LOHC optimization. Different kinds of LOHCs were investigated in terms of a detailed process simulation, techno-economic, and market uncertainty analysis for a 1000 Nm³/h in ref (Brigljević et al., 2020). Lee et al. (Lee et al., 2021). made a comparative energy analysis on LOHCs. This study compared LOHCs to lithium-ion batteries, concentrated solar power, and regenerative hydrogen fuel cells with compressed hydrogen in terms of energy storage technologies, and the advantages/disadvantages were described for different periods. Reactor-based optimization was done in (Rao et al., 2022). Based on the heat exchangers and catalyst at various feed temperatures, reactor wall temperature, and hydrogen burner efficiency, preliminary cost optimization was carried out. Caglar et al. (Caglar et al., 2023). investigated LOHC hydrogen storage for a stand-alone energy application. They analyzed the system in terms of thermo-sustainability, economic and

exergetic methods. They found that LOHC included the system's energy and exergy efficiencies at 44 % and 17 %; the payback period for this application was found to be 3.43.

On the other hand, studies on methanol production show that it is an emerging technology that bridges the gap in the transition from fossil fuels to renewable energy (Roode-Gutzmer et al., 2019). Alsayegh et al. (Alsayegh et al., 2019). examined methanol production with captured CO₂ and photocatalytic water splitting. They found that the plant lifetime is 30 years with 0.060 \$/kWh electricity price and 0.020 \$/kg CO₂ cost, which has a break-even value for methanol at 0.96 \$/kg for the best-case scenario. Based on a target solar to-hydrogen efficiency of 10 % in the photocatalytic reactor, the gross energy efficiency of the process was 6.25 %. Battaglia et al. investigated green methanol production to decarbonize the chemical process industry (Battaglia et al., 2021). They showed the techno-economic feasibility of employing "green" methanol as a chemical vector for the decarbonization of the process industry, even in realistic economic scenarios. Herdem et al. (Herdem et al., 2020). researched the co-generation of power and green methanol. They presented that 103.94 kg methanol per day for 302.74 kg biomass production is possible. Overall, these studies show that renewable methanol production from captured CO₂ has the potential to play an essential role in reducing carbon emissions and achieving sustainable development. Lee et al. (Lee et al., 2020). studied whether renewable methanol production from the captured CO₂ and renewable H₂ is feasible. According to their results, decreasing the renewable H₂ production cost and increasing the CO₂ tax credit for a methanol production capacity of 100 tons per day would make renewable methanol synthesis economically feasible. Ravikumar et al. evaluated the environmental benefits of producing green methanol (Ravikumar et al., 2020). They established 14 scenarios and claimed that the CO₂ avoided by using RE on the grid is more significant than that from producing carbon dioxide capture unit methanol by 660–11960 kg CO₂/ton methanol across the 14 scenarios. Sollai et al. made a techno-economic analysis for green methanol production (Sollai et al., 2023b). They calculated the levelized cost of methanol as 960 €/t (about 175 €/MWh) and concluded that it could be competitive in the mid-future. Bube et al. (Bube et al., 2024). proposed a novel approach for decentralized methanol production from biogas. The investigated production concept offers the opportunity to make sustainable biomass resources available for material use and significantly increases carbon efficiency and production capacity compared to purely biogas-based concepts. Safer et al. (Safer et al., 2024). assessed a multigeneration system to synthesize e-methanol and green hydrogen in terms of techno-economic. The results demonstrate the system's ability to achieve production rates of 430.25 kg/h of e-methanol and 190 kg/h of green H₂. Nimkar et. Al. (Nimkar et al., 2017) analyzed thermally coupled reactors for methanol synthesis using exergy and exergoeconomic methods.

In this paper, a different method to utilize CO₂ and produce methanol, which is a promising fuel and common industrial chemical, is suggested. As remarked in the review of the state of the art, no similar literature considers the key components such as LOHC, CCS, and Methanization Reactor (MTH) used in this research. Moreover, this research introduces a novel approach that integrates the technical-economic cycle and includes environmental aspects. For this purpose, the system's economic, thermo-sustainable, and exergetic performance are investigated. The main objective is to determine the feasibility of this system, its economic parameters, and exergetic performance, as well as evaluate and discuss the results. The main novelty of this paper is to propose a LOHC system as a hydrogen storage mechanism, which has been investigated for energy generation until now. This approach is novel as there is no similar system in the open literature, although some studies about renewable hydrogen and captured CO₂ exist. In these studies, hydrogen is mainly produced by electrolysis, and methanol is produced directly by biomass. In this study, a new and novel method is proposed, and another advantage of this method is that hydrogen can be

produced and stored to be sold or utilized by changing the capacity of the biomass gasifier. However, it is not the subject of this paper. Exergy, exergoeconomic, and fundamental economic analyses have been conducted, and the results are discussed.

2. System description

The system includes a biomass gasifier, a LOHC system (hydrogenation reactor, dehydrogenation reactor, and preheaters), a syngas compressor, a gas turbine cycle, a heat recovery steam generator, a carbon dioxide capture unit, a feed gas compressor, and a methanol reactor. The operation of the process can be described as follows:

Syngas is produced in the biomass gasifier, and the syngas heat is used to meet heat needs in the LOHC system. It is pressurized by the syngas compressor for high pressure demanded by the dehydrogenation reactor. Before the syngas compressor runs, heat is transferred to the water to reduce the syngas temperature and compressor power. Then, some syngas are sent to the dehydrogenation reactor to meet the heat required by the reaction and finally to the hydrogenation reactor to absorb and store hydrogen. The rest of the Syngas is directly sent to a gas turbine, and another part of the Syngas coming from the hydrogenation reactor is mixed with it to generate electricity. A heat recovery steam generator utilizes the high-temperature combusted gas the gas turbine creates to produce steam. Some portion of this steam is used for the CCS plant, and the rest of the steam can be used for some processes such as heating, cooling, and process needs. The captured carbon dioxide and the hydrogen stored in LOHC are mixed in a mixer after the dehydrogenation system releases the hydrogen, and the remaining hydrogen can be sold as a product. This gas is called the methanol reaction feed gas and must be pressurized for the reactor's needs. Before compression, the feed gas temperature must be reduced to decrease the compressor power. After the feed gas compressor, its temperature must be reduced to the temperature of the methanol reactor. Finally, methanol is obtained as a product. This system can provide electricity, heat, and methanol as products, avoiding CO₂ emissions and taxes. The economic analysis of the system is carried out for Spain. A schematic of the system can be seen in Fig. 1.

Assumptions made for this study are listed as follows:

- Steady-state conditions are considered.
- Pressure drops in heat exchangers are neglected.
- Ambient conditions are taken as 25 °C and 1 bar
- Gases are assumed to be ideal
- Specific heats for loaded and unloaded liquid hydrogen carriers are constant
- Tar production in gasification is neglected

3. Modelling

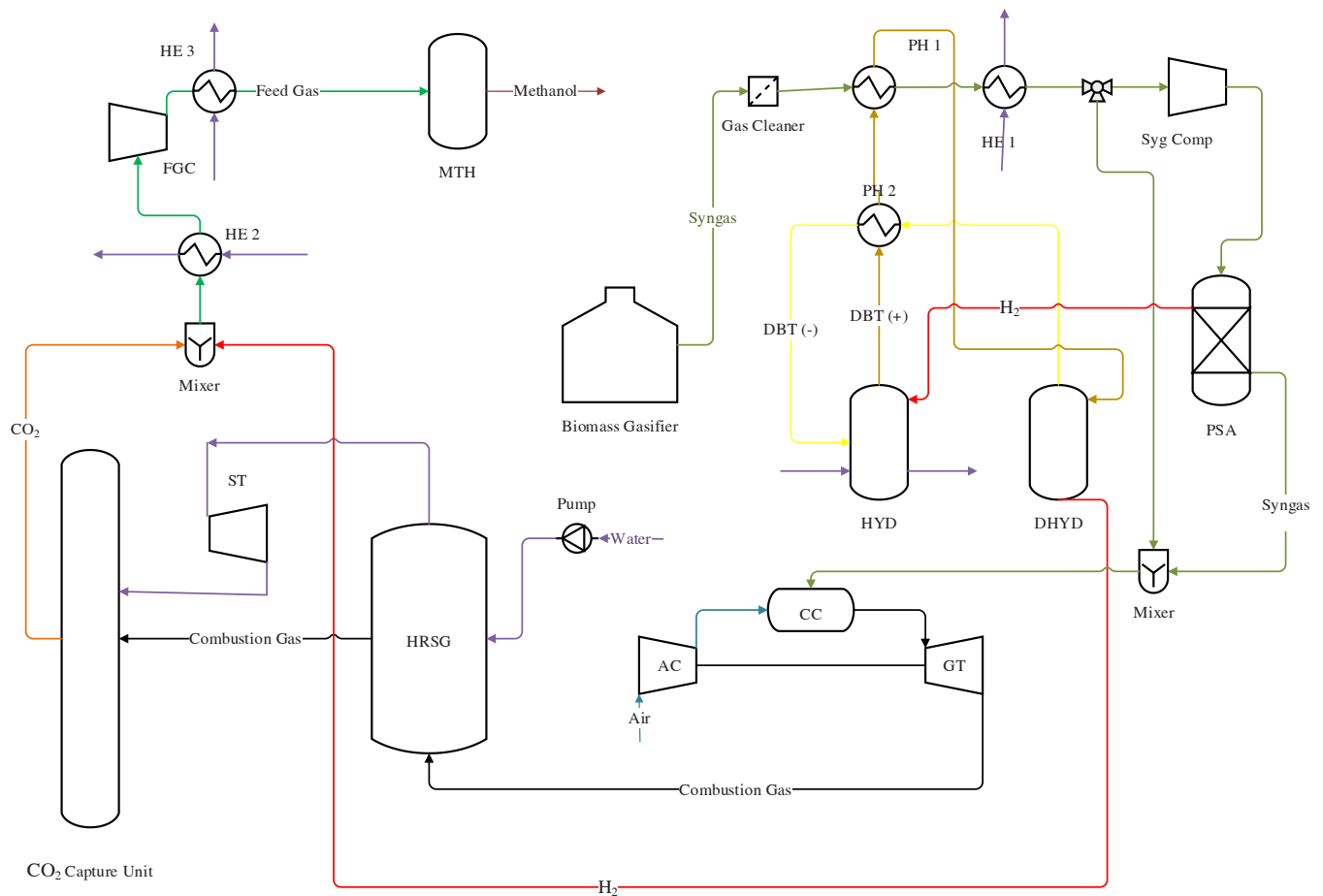
The proposed system is analyzed for exergy, economic, exergoeconomic, and exergetic sustainability indices. All calculations were made in the Engineering Equation Solver (EES). This section presents a detailed description of the calculation method.

3.1. Exergy analysis

Exergy analysis is the most useful thermodynamic tool for determining inefficiencies. Exergy is the reversible work output obtained from a system or process while brought to thermodynamic equilibrium with the environment. This paper assumes that the environmental conditions, temperature, and pressure are 300 K and 100 kPa, respectively. Physical and chemical exergy of any substance are shown in Eq. 1 and Eq. 2 (Cengel and Boles, 2019)

$$\dot{E}x_{phys} = (H_T - H_o) - T_o(S_T - S_o) \quad (1)$$

and



COMPONENTS

FGC: Feed Gas Compressor

MTH: Methanization Reactor

PH 1: Preheater 1

PH 2: Preheater 2

Syg Comp: Syngas Compressor

HYD: Hydrogenation Reactor

DHYD: Dehydrogenation Reactor

CC: Combustion Chamber

AC: Air Compressor

GT: Gas Turbine

HRSG: Heat Recovery Steam Generator

ST: Steam Turbine

PSA: Pressure Swing Adsorber

Fig. 1. Schematic of the system.

$$\dot{E}x_{chem} = \sum_i x_i \varepsilon_i + \underline{RT}_o \sum_i x_i \ln x_i \quad (2)$$

Where T_o is the environment temperature, H is the enthalpy, S is the entropy, x is the mole fraction, ε is the specific chemical exergy, and R is the universal gas constant. The chemical exergies of gas ($\dot{E}_{XQ}$) were calculated by using the standard chemical exergy values listed in [Table 1](#). Total exergy is the sum of the physical and chemical exergies.

The work output is the exergy itself, however, the exergy of heat is calculated as follows (Eq. 3) (Cengel and Boles, 2019).

$$\dot{E}x_Q = Q \left(1 - \frac{T_o}{T} \right) \quad (3)$$

Exergy is not conserved but destroyed, this is called exergy destruction resulting from the entropy generation. Exergy balance for any system or process is shown in Eq. 4 (Cengel and Boles, 2019)

Table 1
Specific chemical exergy (Mojaver et al., 2019).

Compound	Specific chemical exergy (kJ/kmol)
CH ₄	831650
CO ₂	19870
CO	275100
H ₂ O	9500
H ₂	236100

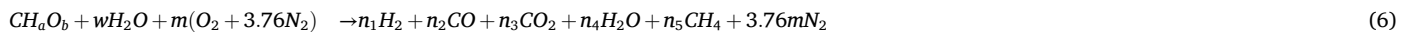
$$\dot{E}x_D = \dot{E}x_F - \dot{E}x_P \quad (4)$$

where, $\dot{E}x_D$ is the exergy destruction, $\dot{E}x_F$ is the fuel exergy which can be defined as exergy input to a system/process to obtain a useful product. $\dot{E}x_P$ is the exergy of the useful product. Exergy efficiency is a very important indicator can be defined as shown in Eq. 5 (Cengel and Boles, 2019).

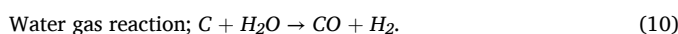
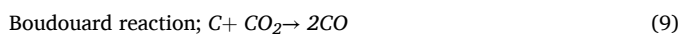
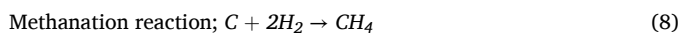
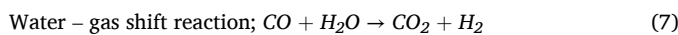
$$\varphi = \frac{\dot{E}x_P}{\dot{E}x_F} = 1 - \frac{\dot{E}x_D}{\dot{E}x_F} \quad (5)$$

3.2. Biomass gasifier

Biomass gasifiers are divided in to the two parts basically as fixed bed and fluidized bed. Fixed bed gasifiers are divided into two sub-categorized downdraft and updraft fixed bed gasifiers. Updraft fixed bed gasifiers are more suitable for direct firing combustion, such as cooking stoves. Downdraft fixed-based gasifiers have lower energy efficiency and are more suitable for small-scale electricity generation with an internal combustion engine (Bocci et al., 2014). Fluidized bed gasifiers have excellent gas-solid mixing rates and thermal inertia; the temperature and solid gas concentration are uniform. Therefore, the biomass conversion is close to 100 %, and outputs are double to ten times higher than fixed beds. Since fluidized bed gasifiers are generally more tolerant and suitable for large installations (Bocci et al., 2014), considering these, a fluidized bed gasification reactor is chosen. In gasification reactors, air, steam, or a mix of them can be used as agents. In this paper, the air is selected as the agent because the CO rate in the syngas must be diluted. The reason is that the catalyzer in the LOHC system is sensitive to CO, which causes catalyzer poisoning. The model of the gasifier was developed using the stoichiometric method. Municipal solid waste was chosen as a biomass feedstock, while the elemental and proximate analyses of the municipal solid waste were taken from Barman et al (Barman et al., 2012). The molecular formula of the biomass was determined by elemental analysis, considering that the biomass only contains carbon, hydrogen, and oxygen. The overall chemical reaction of the gasification is described as follows, Eq. 6:



where n is the mole number. Based on the literature (Barman et al., 2012), the values of a and b were determined as 1.54 and 0.62. The following three equations (Eqs. 7–9) indicate the mass balances used for carbon, hydrogen, and oxygen.



Equilibrium constant relations for the selected reactions are given by

Eqs. (11)–(17) (Barman et al., 2012)

$$n_2 + n_3 + n_5 = 1 \quad (11)$$

$$2n_1 + 2n_4 + 4n_5 = a + 2w \quad (12)$$

$$n_2 + 2n_3 + n_4 = w + 2m + b \quad (13)$$

$$\ln K_1 = \exp \left(-\frac{\bar{g}_{CO}^o}{RT} - \frac{\bar{g}_H^o}{RT} + \frac{\bar{g}_{CO}^o}{RT} + \frac{\bar{g}_H^o}{RT} \right) \quad (14)$$

$$\ln K_2 = \exp \left(-\frac{\bar{g}_{CH}^o}{RT} + 2\frac{\bar{g}_H^o}{RT} \right) \quad (15)$$

$$K_1 = \frac{n_1 \times n_3}{n_2 \times n_4} \quad (16)$$

$$K_2 = \frac{n_5 \times n_{total}}{n_1^2} \quad (17)$$

After determining the number of moles, the enthalpies of the exit gas were calculated using Eqs. 18 and 19 (Barman et al., 2012).

$$c_p(T) = a + bT + cT^2 + dT^3 \quad (18)$$

The coefficients used in Eq. 17 are shown in Table 2 (Cengel and Boles, 2019).

$$h_i(T) = \int_{298.15}^T c_p dT \quad (19)$$

The total enthalpy of the syngas can be calculated as follows (Eq. 20) (Cengel and Boles, 2019)

$$H = \sum_i n_i h_i(T) \quad (20)$$

And the specific entropy of a species can be obtained by Eq. 21 (Cengel and Boles, 2019)

$$s_i(T) = \int_{298.15}^T c_p \frac{dT}{T} \quad (21)$$

Exergy of any substance is determined as a sum of the physical and chemical exergies as follows:

The exergy of the biomass was determined in Eq. 22 (Cengel and Boles, 2019)

$$\dot{E}x_B = \lambda \times \dot{m}_B \times LHV_B \quad (22)$$

where LHV_B is the lower heating value for the biomass, $\dot{m}_B$ is the mass flow rate of the biomass, and λ is the exergetic coefficient for any substance and calculated as shown in Eq. 23 (Barman et al., 2012):

Table 2
Coefficients in Eq. 19 (Cengel and Boles, 2019).

	a	$b \times 10^{-2}$	$c \times 10^{-5}$	$d \times 10^{-9}$	Temperature range (K)
Air	28.11	0.1967	0.4802	-1.966	273–1800
H ₂	29.11	-0.1916	0.4003	-0.8704	273–1800
CO	28.16	0.1675	0.5372	-2.222	273–1800
CO ₂	22.26	5.981	-3.501	-7.469	273–1800
H ₂ O	32.24	0.1923	1.055	-3.595	273–1800
CH ₄	19.89	5.204	1.269	-11.010	273–1500

$$\lambda = \frac{1.404 + 0.0177 \times \left(\frac{H}{C}\right) - 0.3328 \times \left(\frac{O}{C}\right) \left(1 + 0.537 \times \left(\frac{H}{C}\right)\right)}{1 - 0.4021 \times \left(\frac{O}{C}\right)} \quad (23)$$

where C , H , and O are the mass fractions of the compound. Product exergy (or exergetic product) rate is calculated using Eqs. 1–3 for the syngas. Fuel exergy (or exergetic input) to the gasifier is obtained by using Eq. 22. The lower heating value of the biomass was calculated as 20.13 MJ/kg and the gasifier temperature is assumed as 923.15 K.

3.3. LOHC system and syngas compressor

Dibenzyl toluene was chosen as the hydrogen carrier due to its relatively low price, high hydrogenation/dehydrogenation efficiency and high gravimetric storage density (6.2 wt%) (Haupt and Müller, 2017). Since it is used in an industrial LOHC unit as a H_2 carrier (ANON), which makes it an attractive choice for the performance evaluation of the LOHC-containing system. The LOHC unit works based on the hydrogenation and dehydrogenation of LOHC under the right conditions. The operating conditions in hydrogenation and dehydrogenation reactors are 180 °C/50 bar and 310 °C/1 bar respectively (Haupt and Müller, 2017). The hydrogenation process is an exothermic reaction and while the dehydrogenation process is endothermic and the reaction heat is 65.4 kJ/mol H_2 (Li et al., 2023). As mentioned above high purity hydrogen needed in hydrogenation and pressure swing adsorber is included in LOHC system.

3.3.1. Hydrogenation

In the first step, the discharged LOHC mass flow rate (LOHC₋) in the hydrogenation reactor is calculated (Li et al., 2023), Eq. 24:

$$\dot{m}_{LOHC-} = \frac{\dot{m}_{H_2, in} \bullet M_{LOHC-}}{9 \bullet M_{H_2}} \quad (24)$$

here $\dot{m}_{H_2, in}$, M_{LOHC-} , M_{H_2} are hydrogen flow rate entering the hydrogenation reactor, the molecular weight of discharged LOHC (272 kg/kmol) and the molecular weight of the H_2 (2 kg/kmol) (Atsonios et al., 2016).

The heat generated in hydrogenation reactor ($\dot{Q}_{hyd}$) is calculated using Eq. 25 (Li et al., 2023):

$$\dot{Q}_{hyd} = \frac{\dot{m}_{H_2, in} \bullet q}{M_{H_2}} \quad (25)$$

where, q is the reaction heat.

3.3.2. Dehydrogenation

Firstly, the loaded LOHC (LOHC₊) must be calculated and according to conversion of the mass, Eq. 26:

$$\dot{m}_{LOHC+} = \dot{m}_{LOHC-} + \dot{m}_{H_2, in} \quad (26)$$

and hydrogen released from the dehydrogenation reactor is shown in Eq. 27 (Li et al., 2023):

$$\dot{m}_{H_2, out} = \frac{9 \bullet \dot{m}_{LOHC+} \bullet M_{H_2} \bullet \eta_{dhyd}}{M_{LOHC+}} \quad (26)$$

where η_{dhyd} is the dehydrogenation efficiency which is 0.97 (Li et al.,

2023). The heat required by the dehydrogenation is presented in Eq. 28 (Li et al., 2023):

$$\dot{Q}_{dhyd} = \frac{\dot{m}_{H_2, out} \bullet q}{M_{H_2}} \quad (28)$$

3.3.3. Preheaters

For the preheater 1 (PH1), the heat balance can be written as (Eq. 29) (Li et al., 2023)

$$\dot{Q}_{PH1} = \dot{m}_{LOHC+} \bullet c_{p, LOHC+} \bullet (\Delta T_{LOHC+}) = \dot{m}_{syn} \bullet c_{p, syn} \bullet (\Delta T_{syg}) \bullet \beta \quad (29)$$

where $c_{p, LOHC+}$ is the specific heat of the loaded LOHC (449.5 J/kmol) (Li et al., 2023), $\dot{m}_{syn}$, $c_{p, syn}$, β are mass flow rate of the syngas, specific heat of the syngas and the effectiveness of the heat exchanger respectively.

For the preheater 2 (PH2), the heat balance can be written as shown in Eq. 29 (Li et al., 2023)

$$\dot{Q}_{PH2} = \dot{m}_{LOHC+} \bullet c_{p, LOHC+} \bullet (\Delta T_{LOHC+}) \bullet \beta = \dot{m}_{LOHC-} \bullet c_{p, LOHC-} \bullet (\Delta T_{LOHC-}) \quad (30)$$

where $c_{p, LOHC-}$ is the specific heat of the unloaded LOHC (242.3 J/kmol) (Li et al., 2023), and the efficiency of the preheaters is assumed to be 0.85 (Li et al., 2023).

3.3.4. Syngas compressor

A syngas compressor is used to press the syngas to the hydrogenation operation pressure, which is 50 bar. However, after the compressor heat of syngas is used to accomplish the dehydrogenation reaction. Pressure losses in the hydrogenation reactor are neglected. Syngas outlet temperature at the syngas compressor can be calculated by Eq. 31 (Silveira and Tuna, 2003)

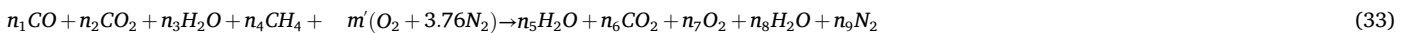
$$T_{out, sygComp} = T_{in, sygComp} \left(1 + \frac{1}{\eta_C} \left((r_{sygComp})^{\frac{k_{syg}-1}{\gamma_{syg}}} - 1 \right) \right) \quad (31)$$

where; $T_{in, sygComp}$, η_C and r_{syg} are the temperature of the syngas at compressor inlet, isentropic efficiency of the compressor and the compression ratio of the syngas compressor. Isentropic efficiency and the pressure ratio of the syngas compressor are 0.75 and 50. Using Table 1, Eq. 17, k values are determined. Work input to the syngas compressor is calculated using Eq. 32 (Cengel and Boles, 2019)

$$\dot{W}_{sygComp} = \dot{m}_{syg} (h_{out, syg} - h_{in, syg}) \quad (32)$$

3.4. Gas turbine

Syngas without H_2 after the hydrogenation reactor is sent to the gas turbine as fuel. The combustion equation is written as (Eq. 33):



For determining the compressor and turbine powers were determined while calculating outlet temperatures from the compressor and turbine is determined using Eqs. 34 and 35 (Silveira and Tuna, 2003).

$$T_{out, AC} = T_{in, AC} \left(1 + \frac{1}{\eta_C} \left((r)^{\frac{k_{air}-1}{k_{ar}}} - 1 \right) \right) \quad (34)$$

$$T_{out,GT} = T_{in,GT} \left(1 - \eta_T \left(\left(r \right)^{\frac{1-k_{gas}}{k_{gas}}} \right) \right) \quad (35)$$

Enthalpy values are determined using outlet temperatures from the compressor and turbine, and finally, compressor and turbine power can be calculated using Eqs. 36 and 37.

$$\dot{W}_{AC} = \dot{m}_{air} (h_{out,AC} - h_{in,AC}) \quad (36)$$

$$\dot{W}_{GT} = \dot{m}_{gas} (h_{in,GT} - h_{out,GT}) \quad (37)$$

Lastly, the net power of the gas turbine is calculated using Eq. 38 (Cengel and Boles, 2019)

$$\dot{W}_{net} = \dot{W}_{GT} - \dot{W}_{AC} \quad (38)$$

Here, $T_{out,AC}$, $T_{in,AC}$, $T_{out,GT}$, $T_{in,GT}$, $\dot{m}_{air}$, $\dot{m}_{gas}$, η_C , η_T , r are the air compressor outlet temperature, the air compressor inlet temperature, the turbine outlet temperature, the turbine inlet temperature, the mass flow rate of the air, the mass flow rate of the commotion gas, the isentropic efficiency of the compressor, the isentropic efficiency of the turbine and pressure ratio, which is the rate of the compressor output pressure to the inlet pressure, respectively. The mass flow rate of the combustion gas can be calculated using the air-fuel ratio (AFR). The values of the parameters used in calculations can be seen in Table 3.

3.5. Heat recovery steam generator

This part produces steam that is used for CCS and to generate power using a steam turbine. The heat produced by the steam is provided by the exhaust gas of the gas turbine. The energy balance equation is Eq. 39 (Li et al., 2023)

$$\begin{aligned} \dot{Q}_{HRSG} &= \dot{m}_{gas} \cdot (h_{out,GT} - h_{out,HRSG}) \cdot \beta \\ &= \dot{m}_{w,HRSG} \cdot (h_{out,w,HRSG} - h_{in,w,HRSG}) \end{aligned} \quad (39)$$

where $h_{out,HRSG}$, $\dot{m}_{w,HRSG}$, $h_{out,w,HRSG}$, $h_{in,w,HRSG}$ are the enthalpy of the combustion gas at heat recovery steam generation (HRSG) outlet, flow rate of the water in the HRSG, enthalpies of the water at the inlet and the outlet of the HRSG.

3.6. Steam turbine

Power is generated by the steam turbine from the steam produced in HRSG. A portion of the steam after the steam turbine, which is at 200 °C and 120 kPa, is sent to CCS's reboiler to meet process requirements. The rest of the steam is a heat source for different applications. Power generated in the steam turbine is calculated as shown in Eq. 40 (Cengel and Boles, 2019).

$$\dot{W}_{st} = \dot{m}_{w,HRSG} \cdot (h_{out,w,HRSG} - h_{out,w,ST}) \quad (40)$$

$h_{out,w,ST}$ is the enthalpy of the steam at the outlet of the steam turbine.

3.7. Carbon dioxide capture system

After the HRSG, combusted gas is sent to CCS to capture carbon dioxide. The CO₂ capture efficiency is 90 % with a 30 %

Table 3
Assumed parameters for gas turbine calculations (Cengel and Boles, 2019).

Parameter	Unit	Value
$T_{in,T}$	K	1300
η_C	-	0.75
η_T	-	0.75
r	-	8
AFR	-	50

monoethanolamine (MEA) solution, and the energy needed for the amine regeneration is 3.46 MJ/kgCO₂ (Brigljević et al., 2020).

3.8. Methanol reactor and the feed gas compressor

Feed gas is compressed to the methanol reactor pressure in the feed gas compressor. The temperature at the out of the feed gas compressor is shown in Eq. 41 (Silveira and Tuna, 2003)

$$T_{out,FGC} = T_{in,FGC} \left(1 + \frac{1}{\eta_C} \left((r_{sFGC})^{\frac{k_{FGC}-1}{k_{FGC}}} - 1 \right) \right) \quad (41)$$

Power input to the feed gas compressor is (Eq. 41) (Cengel and Boles, 2019).

$$\dot{W}_{FGC} = \dot{m}_{FG} \cdot (h_{out,FGC} - h_{in,FGC}) \quad (42)$$

where; $T_{in,FGC}$, η_C and r_{FGC} are the temperature of the feed gas at compressor inlet, isentropic efficiency of the compressor and the compression ratio of the feed gas compressor. Isentropic efficiency and the pressure ratio of the syngas compressor are 0.75 and 65.

The final process is the methanol production. In the reactor H₂/CO₂, the total heat demands for crude methanol preheating and the distillation column reboiler demands are 397.4 kWh/tnMeOH. Gas cooling can cover this heat after the reactor; thus, no additional heating system is required (Atsonios et al., 2016). Compression and pumping needs in the methanol reactor are 34.5 kWh/tnMeOH, and the mole flow of the feed gas is 4.05 kmol/kmol_{MeOH} (Atsonios et al., 2016). Exergetic relations of components are presented in Table 4.

3.9. Heat exchangers

The heat exchangers provide heat as steam, which is 120 °C and 100 kPa. Energy balances for HE1, HE2, and HE3 can be written as shown in Eqs. 42–44 (Li et al., 2023).

$$\dot{Q}_{HE1} = \dot{m}_{syg} \cdot c_{p,syg} \cdot (\Delta T_{syg,HE1}) \quad \bullet \quad \beta = \dot{m}_{w,HE1} \cdot c_{p,wHE1} \cdot (\Delta T_{w,HE1}) \quad (42)$$

$$\dot{Q}_{HE2} = \dot{m}_{FG} \cdot c_{p,FG} \cdot (\Delta T_{fg,HE2}) \quad \bullet \quad \beta = \dot{m}_{w,HE2} \cdot c_{p,wHE2} \cdot (\Delta T_{w,HE2}) \quad (43)$$

$$\dot{Q}_{HE3} = \dot{m}_{FG} \cdot c_{p,FG} \cdot (\Delta T_{FG,HE3}) \quad \bullet \quad \beta = \dot{m}_{w,HE3} \cdot c_{p,wHE3} \cdot (\Delta T_{wHE3}) \quad (44)$$

where, $\dot{m}_{FG}$, $\dot{m}_{w,HE1}$, $\dot{m}_{w,HE2}$, $\dot{m}_{w,HE3}$, $\dot{m}_{w,HE3}$ are the mass flow rates of feed gas, water flow rates of the water in heat exchangers, respectively. $c_{p,FG}$, $c_{p,wHE1}$, $c_{p,wHE2}$, $c_{p,wHE3}$ are the specific heat of the feed gas and water in heat exchangers respectively. Effectiveness (β) values are assumed 0.85 (Li et al., 2023).

Table 4
Exergy relations in components.

	Ex_F (kW)	Ex_P (kW)
Gasifier	$\dot{E}x_B + \dot{E}x_{air}$	$\dot{E}x_{chem,syg} + \dot{E}x_{phys,syg}$
LOHC	$\dot{E}x_{H_2,in} + \dot{E}x_{Q_{lyd}} + \dot{W}_{syg,Comp} + \dot{E}x_{Q_{rln}}$	$\dot{E}x_{H_2,out} + \dot{E}x_{Q_{lyd}}$
GT	$\dot{E}x_{chem,sygoH_2}$	$\dot{W}_{net}$
Pump	$\dot{W}_{pump}$	$\dot{E}x_{out,w,Pump} - \dot{E}x_{in,w,Pump}$
HRSG	$\dot{E}x_{out,gas,HRSG} - \dot{E}x_{in,gas,HRSG}$	$\dot{E}x_{out,w,HRSG} - \dot{E}x_{in,w,HRSG}$
Syg. Comp	$\dot{W}_{syg,Comp}$	$\dot{E}x_{out,syg,sygComp} - \dot{E}x_{in,syg,sygComp}$
FGC	$\dot{W}_{FGC}$	$\dot{E}x_{out,FG,FGC} - \dot{E}x_{in,FG,FGC}$
HE1	$\dot{E}x_{out,syg,HE1} - \dot{E}x_{in,syg,HE2}$	$\dot{E}x_{out,w,HE1} - \dot{E}x_{in,w,HE1}$
HE2	$\dot{E}x_{out,FG,HE2} - \dot{E}x_{in,FG,HE2}$	$\dot{E}x_{out,w,HE2} - \dot{E}x_{in,w,HE2}$
HE3	$\dot{E}x_{out,FG,HE3} - \dot{E}x_{in,FG,HE3}$	$\dot{E}x_{out,w,HE3} - \dot{E}x_{in,w,HE3}$
ST	$\dot{E}x_{in,w,ST} - \dot{E}x_{out,w,ST}$	$\dot{W}_{st}$
CCS	$\dot{E}x_{Q,reboiler} + \dot{W}_{need}$	$\dot{E}x_{chem,CO_2}$
MTHR	$\dot{E}x_{chem,H_2} + \dot{E}x_{chem,CO_2} + \dot{W}_{need}$	$\dot{E}x_{chem,methanol}$
System	$\dot{E}x_B$	$\dot{E}x_{Q_{ys}} + \dot{E}x_{chem,methanol} + \dot{W}_{out}$

Table 5

Capital cost correlations (Eypasch et al., 2017; Naserian et al., 2016; Bejan et al., 1996; Spallina et al., 2016; Cavalcanti, 2017; Hosseinpour et al., 2020; Bellotti et al., 2019; Roya et al., 2020). (\$/€ parity is assumed as 0.9).

Component	Cost Correlation for Capital Investment*
Hydrogenation Reactor (Reference year 2017)	$5057.29 \cdot P_0.566_{H_2, \max H_2, \max} \cdot (\text{€})$
Dehydrogenation Reactor (Reference year 2017)	$6888.83 \cdot P_0.576_{H_2, \max H_2, \max} \cdot (\text{€})$
Gas Turbine (Reference year 1996)	$\left(\frac{266.3 \cdot \dot{m}_g}{0.97 - \eta_t} \right) \cdot \ln \left(\frac{P_i}{P_o} \right) \cdot (1 + \exp(0.036T_i - 54.4))$
Combustion Chamber (Reference year 1996)	$\left(\frac{46.08 \cdot \dot{m}_a}{0.995 - \frac{P_o}{P_i}} \right) \cdot (1 + \exp(0.018T_i - 26.4))$
Compressor (Reference year 1996)	$\left(\frac{39.5 \cdot \dot{m}_a}{0.95 - \eta_c} \right) \cdot \ln \left(\frac{P_i}{P_o} \right) \cdot \left(\frac{P_i}{P_o} \right) \cdot (\text{\$})$
HRSG (Reference year 1996)	$6570 \left(\frac{\dot{Q}_{HRSG}}{\Delta T_{LMTD, HRSG}} \right)^{0.8} + 21276 \dot{m}_{steam} + 1184.4 \dot{m}_{gas}^{1.2}$ (\\$)
Pump (Reference year 1996)	$2100 \cdot \left(\frac{\dot{W}}{10} \right)^{0.26} \cdot \left(\frac{1 - \eta}{\eta} \right)^{0.5} \cdot (\text{\$})$
Steam Turbine (Reference year 1996)	$2100 \cdot (\dot{W})^{0.7} \cdot (\text{\$})$
Heat Exchanger (Reference year 2010)	$Z_{HE} = 130 \times \left(\frac{A}{0.093} \right)^{0.78} \cdot (\text{\$})$
Biomass Gasifier (Reference year 2005)	$1600 \times \dot{m}_{syn} \times 0.9 \times 3600^{0.67} \cdot (\text{\$})$
Carbon dioxide Capture (Reference year 2019)	$4.1811 \times 10^6 \times (m_{CO_2})^{0.7} \cdot (\text{\$})$
Methanol Production Reactor (Reference year 2019)	$20.4 \times 10^3 \times (m_{MeOH})^{0.65} \cdot (\text{\$})$
Pressure Swing Adsorber (Reference year 2007)	$27.96 \times 10^6 \times \left(\frac{\dot{m}_{gas}}{17069} \right)^{0.6} \cdot (\text{€})$

Table 6

The fixed costs details (Tian et al., 2023).

Direct costs (DC)	
Cost of investment	$\sum Z_k$
Electrical equipment and materials	$0.13 \sum Z_k$
Piping	$0.33 \sum Z_k$
Purchased-equipment installation	$0.35 \sum Z_k$
Civil, structural, and architectural work	$0.21 \sum Z_k$
Land	$0.05 \sum Z_k$
Service facilities	$0.35 \sum Z_k$
Indirect cost (IC)	
Construction costs	$0.15 \sum DC$
Contingencies	$0.15(1.23 \sum DC)$
Engineering and supervision	$0.08 \sum DC$

3.10. Economic and exergoeconomic analysis

Exergoeconomic is a method that combines the exergy flows and equipment costs. A method called EXCEM (EXergy-Cost-Energy-Mass) is chosen as exergoeconomic method. This method calculates the capital costs since the parameter R (kW/€) is described as the ratio of the exergy destruction rate to capital cost. R can be described exergetic point of view as presented in Eq. 46 (Nimkar et al., 2017)

$$R_{EX,k} = \frac{Ex_{D,k}}{Z_k} \quad (46)$$

Z_k is the capital investment of the k component, and OMC is the Operation and Maintenance Cost, which is assumed to be 10 %. Capital costs are obtained by using some correlations from the literature, which are listed in Table 5.

For determining updated cost, it can be calculated by using Eq. (46) where CEPCI is Chemical Engineering Plant Cost Index

$$Z_{k,2022} = Z_{k,reference} \cdot \frac{(CEPCI_{2022})}{(CEPCI_{Reference})} \quad (47)$$

Other costs, including direct and indirect costs, are calculated as seen in Table 6.

The payback period (PP) is essential for investment decisions in engineering applications. This study uses PP to evaluate the system's economic performance. PP is calculated based on the annual depreciation cost and income (ANON, 2022).

$$PP = \frac{\ln \left(1 - \frac{Z_{total}}{C_{IM}} (I - In) \right)}{\ln \left(\frac{1+In}{1+I} \right)} \quad (48)$$

where I and In the interest rate and inflation rate (the average interest and inflation rates for Spain for 2022 are 3.632 % and 7.88 % (Banco de España, 2023)) and N is the lifetime, which is assumed as 20 years,

The income is calculated considering that electricity is sold to the grid and methanol source, avoiding CO₂ taxes. CO₂, electricity, and methanol prices are 80.63 €/ton, 203.249 €/MWh, and 490\$/ton, respectively (EU ETS, 2023; OMIE, 2022; Wang et al., 2022). The income is calculated by Eq. 49:

$$C_{IM} = P_E \cdot c_E + m_{CO_2} \cdot c_{CO_2} + m_M \cdot c_M \quad (49)$$

here, P_E is the power, c_E is the electricity price, c_{CO_2} is the carbon dioxide price, c_M is the methanol price.

3.11. Exergetic sustainability indexes

In the literature, exergy is used to show the sustainability of any system by utilizing the concept of exergy destruction. In this method, a depletion ratio concept (DR) measures how much exergy source is depleted to produce the desired exergy product. It shows the system's depleted exergy source. It is an implicit indication of the system's environmental performance, i.e., the lower exergy depletion suggests the lower emissions released into the environment. It is calculated as (Rosen et al., 2008) shown in Eq. 50:

$$DR = \frac{\dot{Ex}_D}{\dot{Ex}_F} \quad (50)$$

The inverse of the depletion ratio gives a sustainable index, which is shown as follows (Rosen et al., 2008), Eq. 50:

$$SI = \frac{1}{DR} = \frac{1}{1 - \varphi} \quad (51)$$

Another one is the exergetic ecological index (EECI), obtained by removing the depletion ratio from the exergy efficiency. If this index has a negative value, the depletion of the exergy source is more significant than the product exergy. In other words, the depleted exergy source is higher than the useful part of the exergy source, and the considered system/component must be improved or optimized. product or desired exergy. The lower value of this index suggests that the magnitude depletion of the exergy source is relatively high compared to the exergy of useful product (Açıkkalp and Ahmadi, 2018), Eq. 52:

$$EECI = \frac{E_P - E_D}{E_F} = \varphi - DR \quad (52)$$

However, these methods are considered only exergy-based and provide some results about efficient fuel usage. Different analysis methods should be used for deep sustainability analysis, and these methods are the subject of another paper.

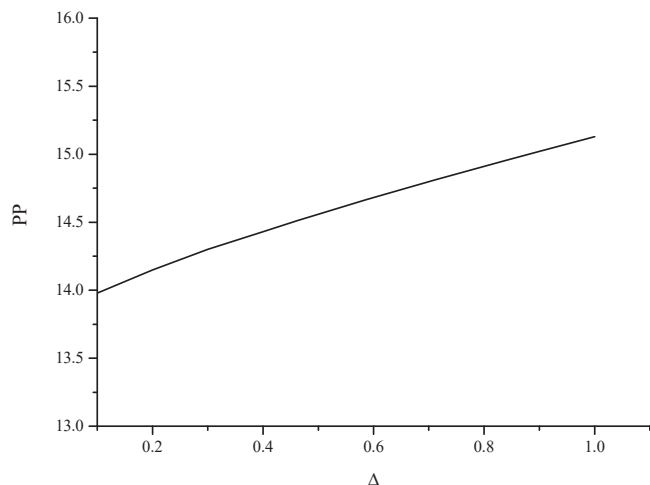


Fig. 2. Effect of Δ on the payback period.

4. Results and discussion

4.1. Effect of hydrogen portion stored in LOHC on the system

This section will analyze the effect of the hydrogen storage ratio in the LOHC system (Δ). The results are shown in Figs. 2–7. Δ is chosen between 0.1 and 1. In reality, storing all hydrogen causes some problems with combustion in the gas turbine cycle; however, optimizing combustion parameters is beyond the scope of this paper.

As seen in Fig. 2–7, Δ has significant and positive effects on all components and the system itself, except the payback period (PP). According to Fig. 2., the payback period increases with Δ , ranging from 13.980 to 15.130. However, this rate of increase in PP can be tolerated since it is about 13 years, which can be assumed as a long value for any engineering application, and these investments need some support and government subsidies. For the thermodynamic and sustainability parameters, including exergy efficiency, depletion ratio, sustainability ratio, and energetic ecological index, Δ has a very effective effect on them. Because they mainly increase or stay stable. For the R_{EX} values, they have decreasing trend, which means that capital costs decrease due to exergy destruction, except for LOHC and METH. The reason the R_{EX}

values of LOHC and METH increase is that their capacity, therefore, the capital costs increase with Δ . As can be seen, hydrogen stored in LOHC for methanol production has mostly a positive effect on the system.

4.2. Exergy analysis and exergetic sustainability indices

As explained in Section 3, exergy analysis is one of the most important thermodynamics performance analyses by determining improved inefficiencies. In this analysis, the exergy destruction rate shows irreversibilities in any system causing inefficiencies and exergy efficiency, which measures the quality of the energy use. The quality of the energy is how close any system/process is to the ideal process, i.e., the Carnot cycle for the power generation processes, and the closer the ideal process is, the lower the inefficiencies. Exergy, exergy-based sustainability indices, and exergoeconomic values are shown in Table 7 for the $\Delta=0.5$, which is selected as an average value. For the EECI value it is essential to remark that if this index has a negative value, the depletion of the exergy source is bigger than the product exergy. This means that the depleted exergy source is higher than the useful part of the exergy source, and the considered system/component must be improved or optimized. product or desired exergy.

Results show that the maximum exergy efficiency is 91.1 % in the pump, and the minimum efficiency is 6.5 % in the gasifier, which means irreversibilities or entropy generation is very high because of the chemical reactions in the gasifier. Since the entropy generation rate in chemical reactions is generally higher, Fig. 8. When Fig. 8 is investigated, it can be seen that MTH has an exergy efficiency of more than 80 %, Pump has an exergy efficiency of more than 90 %, exergy efficiency of the CCS is 17 % and 9 % for the HRSG which has the second-lowest efficiency. So, among the main components, except the gasifier, HRSG, and carbon dioxide capture system, have lower exergy efficiencies and must be focused on improving. Lower exergy efficiency values of these components are connected to higher values of the DR, and this index is strictly related to exergy efficiency, as is seen in Eq. (5). So, the higher values of the DR mean the lower values of the exergy efficiencies. It is an expected result since DR measures the depleted exergy source; in other words, it is the portion of the exergy sources that cannot be converted into useful exergy output and cannot be utilized later. The DR values of the Gasifier, CCS, and HRSG are 93.5 %, 83.2 %, and 91 %, respectively, showing that the exergy source's depletion is nearly 85 % on average. Finally, DR values are reversibly proportional to the exergy efficiency.

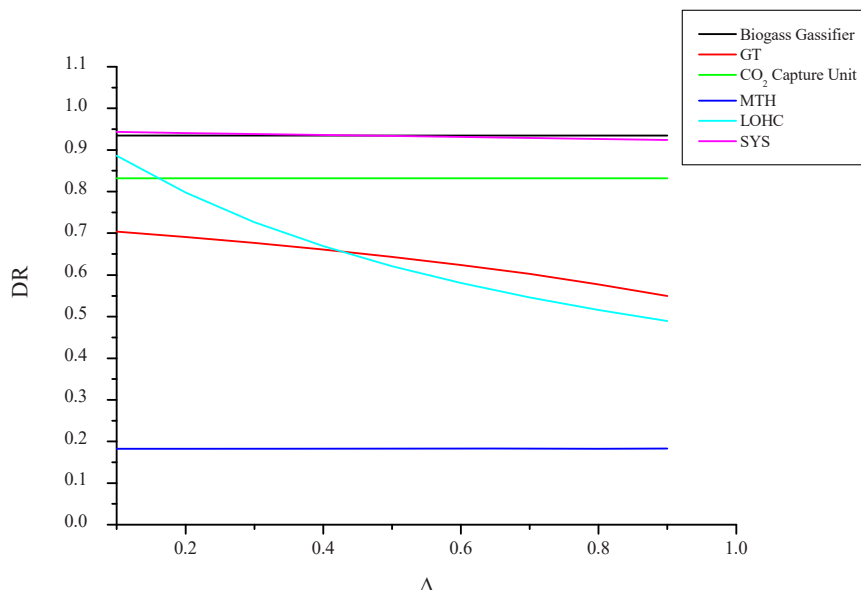
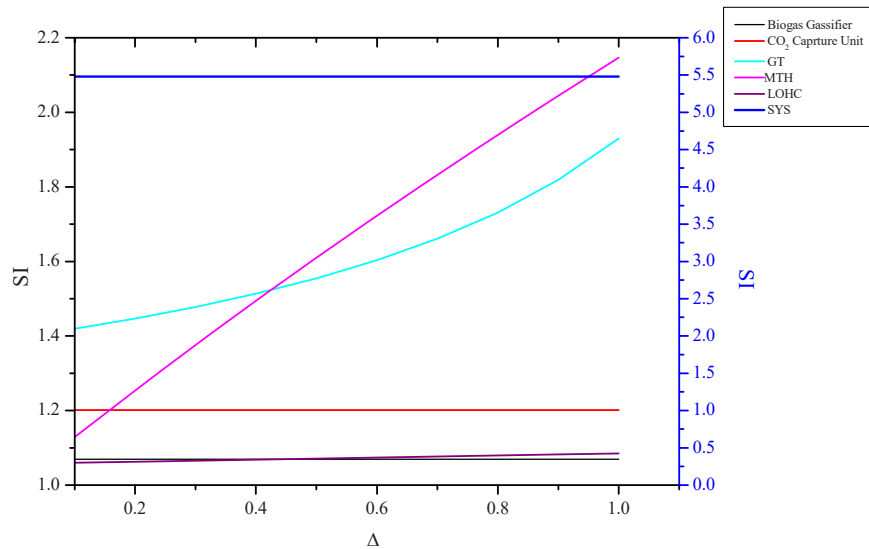
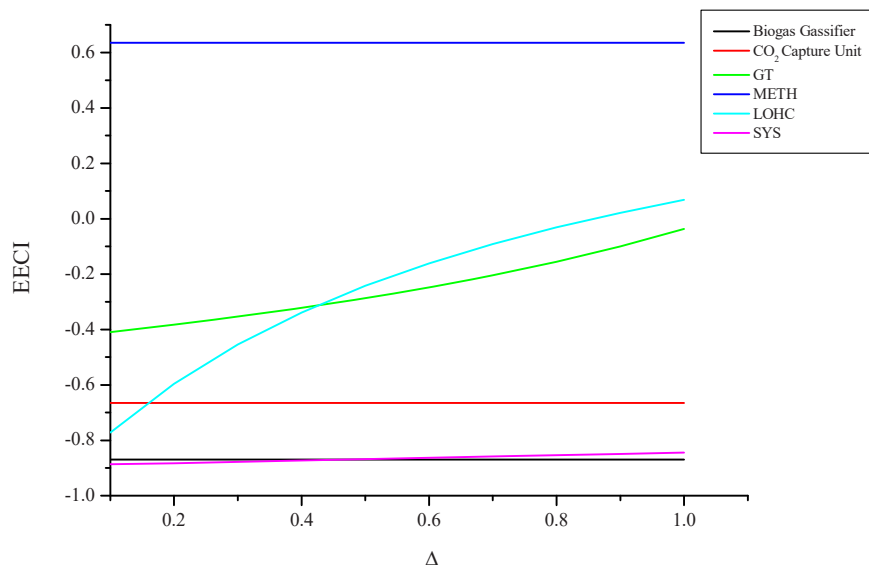


Fig. 3. Effect of Δ on the DR.

Fig. 4. Effect of Δ on the SI.Fig. 5. Effect of Δ on the EECl.

SI value represents how the exergy source is utilized as efficiently. It is the reverse of the DR, and this value is strictly connected with exergy efficiency, as seen in Eq. (51). The pump is the most sustainable component since its SI is 11.284, respectively. The higher value means a more sustainable system, and if the SI value is close to the one, then the system is said to be sustainable. According to these, gasifiers, HRSG, and CCS are not sustainable components of the proposed system. However, SI is an indicator for considering the environmental impact of any system/process, and it does not include other tools of sustainability, which are economic, social, and material consumption. Values for the DR and SI are illustrated in Fig. 9 and Fig. 10. The final exergy-based sustainability index is the EECl. This index provides a better insight and comparison to useful and depleted exergy output. As shown in Eq. (52), EECl is calculated by removing DR from the exergy efficiency, which is equal to the difference between the useful exergy and depleted exergy. If this index has a negative value, it means that the depleted exergy in the system is more significant than the useful exergy output. EECl values for the proposed system are presented in Table 7 and Fig. 11.

According to the results, gasifier, HRSG, FGC, HE1, HE3 and CCS

have negative values. Results show that depleted exergy is more significant than the useful exergy output, which means the exergy destruction rate resulting from the inefficiencies and the source of the inefficiencies are more significant than the useful exergy. In other words, the exergy source is depleted more than converted to a useful product. For the whole system, the exergy efficiency of the system is calculated as 0.066, which means that more than 6 % of the exergy sources are converted into a useful product. The DR and SI values are 0.934 and 1.071, respectively. According to these results, it can be said that its exergy destruction rate is high; however, it can be negligible. Since it is a renewable energy system, there is no environmental emission. In addition, captured carbon dioxide is utilized for a product. The EECl value is found to be 0.237. This positive value means that the portion of the product exergy is less than the depleted exergy.

As can be seen from the equations, the exergy destruction value is responsible for inefficiencies, and chemical reactions cause huge exergy destructions because of their irreversible nature. The system considered in this paper includes many chemical processes, and these processes create the most thermodynamic-based inefficiencies.

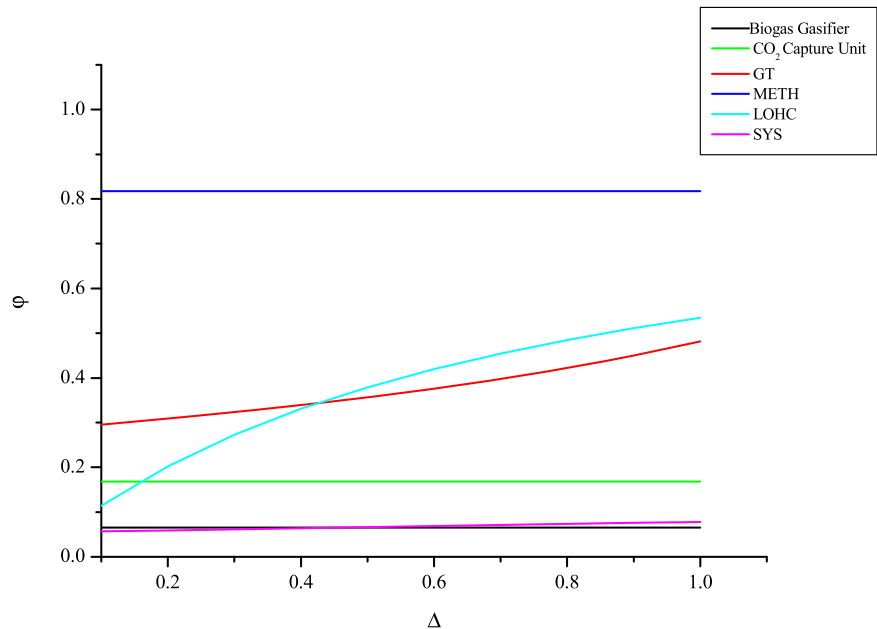


Fig. 6. Effect of Δ on the exergy efficiencies.

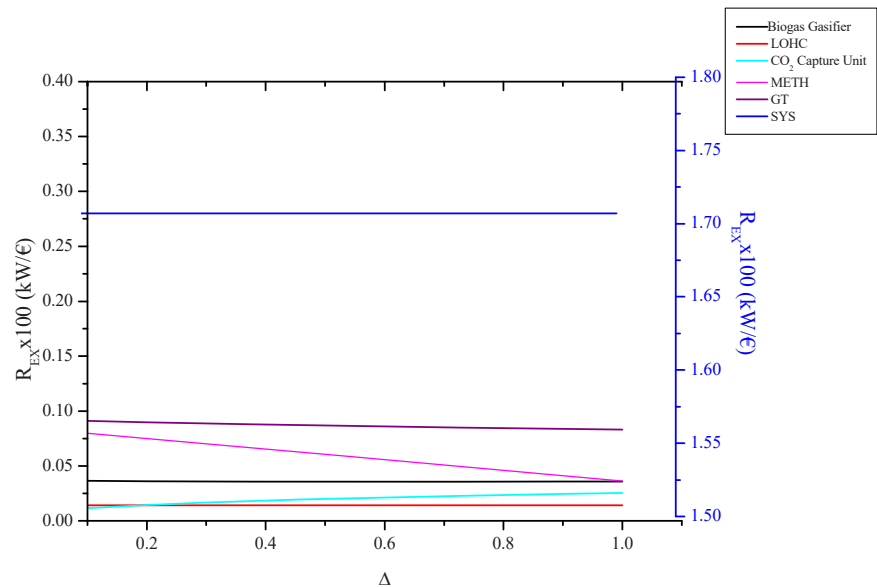


Fig. 7. Effect of Δ on the REX.

Table 7
Results of Exergy Based Analysis (Δ=0.5).

Components	Ex_F (kW)	Ex_P (kW)	Ex_D (kW)	φ	$EECI$	DR	SI	Z (€)	$R \times 100$ (kW/€)
Gasifier	62997	4108	58889	0.065	-0.870	0.935	1.070	3450000	1.707
LOHC	2365	895.700	1469.300	0.379	-0.243	0.621	1.610	4467596	0.036
GT	4250	1516	2734.000	0.357	-0.287	0.643	1.554	4513000	0.061
Pump	8.542	7.785	0.757	0.911	0.823	0.089	11.284	2566	0.030
HRSG	38428	3467	34961	0.090	-0.820	0.910	1.099	678420	5.153
Syg. Comp	1354	687.500	666.500	0.508	0.016	0.492	2.032	240915	0.277
FGC	47.460	22.980	24.480	0.484	-0.032	0.516	1.939	13242	0.185
HE1	1083	293.600	789.400	0.271	-0.458	0.729	1.372	25512	3.094
HE2	2.485	1.791	0.694	0.721	0.441	0.279	3.581	25512	0.003
HE3	19.160	5.068	14.092	0.265	-0.471	0.735	1.360	25512	0.055
ST	3474	1957	1517	0.564	0.127	0.436	1.773	865587	0.175
CCS	1486	249.200	1236.800	0.168	-0.665	0.832	1.201	8167000	0.015
MTHR	927.300	758.100	169.200	0.818	0.635	0.182	5.480	844285	0.020

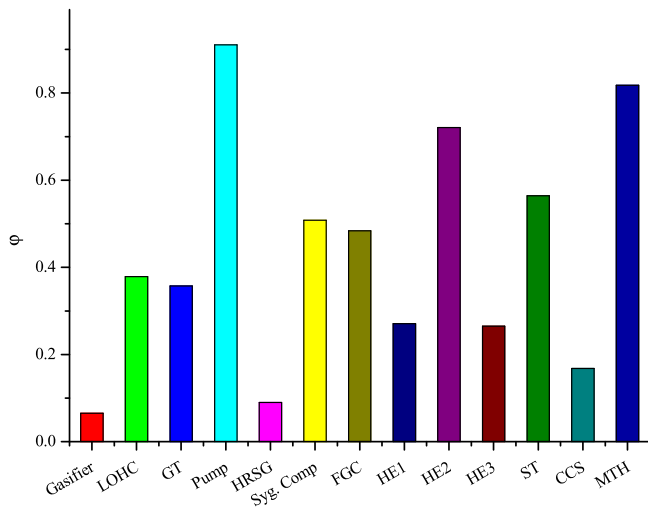


Fig. 8. Exergy efficiencies (Δ=0.5).

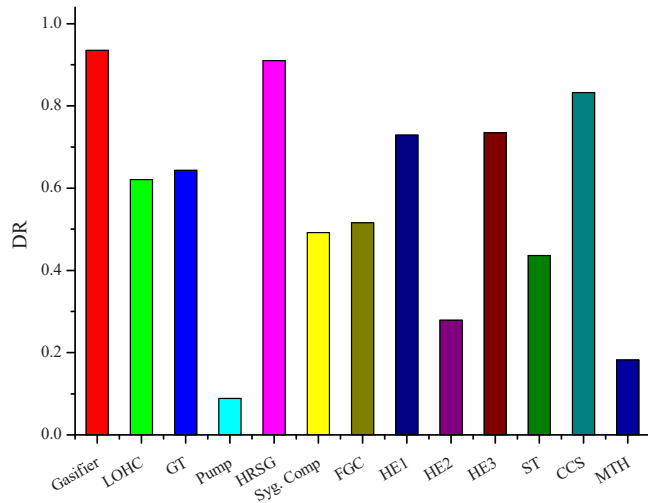


Fig. 9. Depletion Ratios (Δ=0.5).

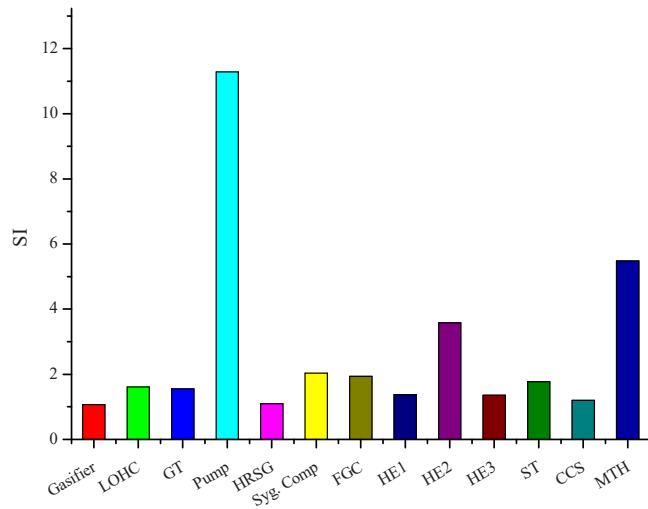


Fig. 10. Sustainability Indexes (Δ=0.5).

4.3. Economic and exergoeconomic evaluations

Basic economic parameters are calculated and evaluated firstly which are C_{IM} and payback period (PP). Corresponding values are found as, 3737000 €/year and 14.56 years for $\Delta=0.5$. 14.56 years are assumed as long for engineering application, however, industrial applications of the LOHC, CCS and MTH are relatively new and still continuing to research. In the near future, it can be expected that their costs will decrease and these kinds of systems will be more feasible.

Exergoeconomic analysis is a method combining thermodynamic and economic methods. This paper chooses EXCEM analysis as exergoeconomic analysis, which can be described as the cost of any component's exergy destructions or inefficiencies (R_{EX}). According to the results in Table 7, the R_{EX} value is the maximum for the HRSG as 5.153×10^{-2} (kW/€). Other ones are 1.707×10^{-2} (kW/€) for the gasifier and 3.094×10^{-2} (kW/€) for the HE1. In other words, R_{EX} shows monetary expenses due to irreversibility (lost work) or depletion of exergy sources. An interesting result was obtained for the MTH and CCS. Exergetic parameters and exergetic-based sustainability indices show very negative results for these components. However, their R_{EX} values are equal to 0.020×10^{-2} (kW/€) and 0.015×10^{-2} (kW/€), respectively. Comparing these values to the other three components, which have the maximum R values, are lower dramatically. The reason for this is that MTH and CCS have high capital investment costs as well as high exergy destruction rates. For the whole system, R_{EX} equals 0.499×10^{-2} (kW/€), which can be assumed to be relatively low compared to HRSG, gasifier, and HE1. The main results for the analysis are synthesized in Table 8.

4.4. Validation of the proposed system

According to the authors' knowledge, there is no similar research in the literature considering some key components such as LOHC, CCS, and MTH.

The exergy efficiency of the LOHC is compared to the results in (Li et al., 2023). In this research, the exergy efficiency of the LOHC for different variations, including fuel cell or industrial chain, was presented deeply. According to the results, exergy efficiencies range from 0.423 to 0.77 for different configurations. In our research, the exergy efficiency of the LOHC ranged from 0.114 to 0.534, which is consistent with the ref (Li et al., 2023).

In ref (Sanchez and de Oliveira, 2015), Sanchez and de Oliveira Jr. found the exergy efficiency of the CCS integrated into an offshore petroleum plant to be 0.085. In ref (Ali et al., 2023), Ali et al. calculated the CCS unit with MEA to be 0.123. In this paper, the exergy efficiency of the CCS is obtained as 0.168, and these results are consistent with those in the literature.

In ref (Nimkar et al., 2017), different types of methanol reactors were considered using the exergy approach. In this paper, exergy efficiencies range from 0.316 to 0.911, and the MTH's exergy efficiency is determined as 0.818. As seen, this value falls into a corresponding range of values. These results obtained from the literature are recalculated by using our method. For the LOHC, because there are many scenarios presented in ref (Li et al., 2023), only an electrolyzer-solid oxide fuel cell has been chosen for the validation. Similarly, a membrane coupled reactor was chosen among several reactor types in ref (Nimkar et al., 2017). Results are summarized in Table 9.

5. Conclusions

A new biomass-based CHP system is proposed in this study. LOHC is used to store and release hydrogen to produce methanol from captured carbon dioxide. Some important results and recommendations are listed below:

- The proposed system's exergy efficiency is 6.6 %, which means more than 6 % of the fuel exergy (exergy of the biomass) is used as useful

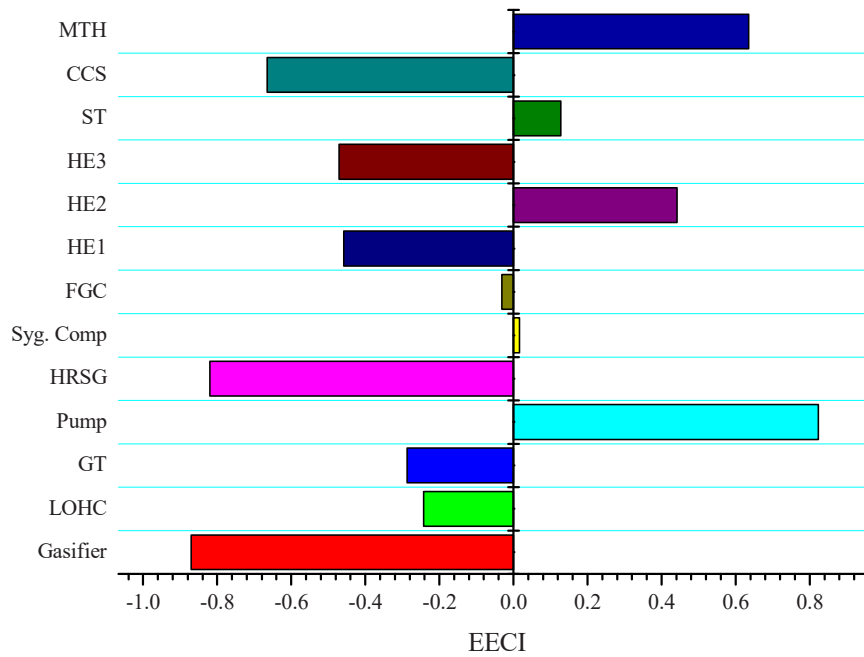


Fig. 11. Exergetic Ecological Indexes ($\Delta=0.5$).

Table 8

Main results for the analysis.

Exergy efficiency	6.6 %		
Exergy based sustainability	DR: 0.0934	SI: 1.071	EECI: -0.868
Payback period	$\Delta=0.1$: 13.980 years	$\Delta=0.5$: 14.56 years	$\Delta=1$: 15.130 years
Exergoeconomic analysis: R_{EX} (kW/€)	Global: 0.087×10^{-2}		
Exergoeconomic analysis: R_{EX} (kW/€)	HRSG: 5.153×10^{-2}	Gasifier: 1.707×10^{-2}	HE1: 3.094×10^{-2}

Table 9

Recalculation of literature results with the method in this study.

	Literature Results	This Study
LOHC	0.611 (Sollai et al., 2023b)	0.642
CCS	0.085 (Bellotti et al., 2019) and 0.123 (Roya et al., 2020)	0.097 and 0.118
MTH	0.864 (Caglar et al., 2023)	0.862

- product exergy. The reason of this small exergy efficiency is high entropy generation rates in chemical reactions.
- The exergy-based sustainability assessment is performed using indices such as DR, SI, and EECI, which are equal to 0.934, 1.071, and -0.868 , respectively. DR shows us that 93.4 % of the exergy source is depleted Exergetic Sustainability index is 1.071, it means that it is slightly higher than 1, which is the lowest border of the sustainability index. So, the proposed system can be considered relatively sustainable in terms of exergetically. EECI provides a comparison between useful products and depleted exergy. This index is desired to be as high as possible. A negative value means depleted exergy is more than useful output.
 - According to exergy efficiency, DR, and SI values, the most problematic components are the gasifier, HRSG, and CCS. However, EECI shows us FGC and heat exchangers should be improved.
 - Basic economic analysis shows that the payback period for $\Delta=0.5$. 14.56 years is assumed to be long for engineering applications;

however, industrial applications of the LOHC, CCS, and MTH are relatively new and still being researched.

- Exergoeconomic analysis reveals that REX value of the system is 0.087×10^{-2} , while HRSG, gasifier and HE1 are obtained as 5.153×10^{-2} (kW/€), 1.707×10^{-2} (kW/€), 3.094×10^{-2} (kW/€) respectively.
- Results are pointed out that HRSG, heat exchangers, and gasifier must be focused on in terms of exergy and exergoeconomic approaches.

Although, the whole system has not good exergetic indicators and not feasible for now, exergoeconomic parameters presents investment depletion is satisfying. However, in the near future, it might be possible that the proposed system has better economic and energetic indicators along with development in LOHC, CCS and MTH. It should be taken into account that the proposed system is renewable energy sourced and there is no carbon dioxide emission to the environment which saves money avoiding carbon tax. Besides, methanol that can be potentially used as fuel or sold for additional economical input is produced. Another alternative investigation for the future studies is to use methanol for hydrogen storage and LOHC system can be removed and methanol reformer reactor can be added. In this way, LOHC costs can be eliminated. In addition, further detailed sustainability methods like extended-exergy and Thermoecological cost should be applied in the next studies for different variations of the proposed system. It is recommended that the proposed system be improved and similar variations investigated.

CRediT authorship contribution statement

David Borge-Diez: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation. **Emin Açıkkalp:** Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Demet Canpolat Tosun:** Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Rustem Zairov:** Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Ana I. Palmero-Marrero: Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Onder Altuntas:** Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

All the data are presented in the paper itself.

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